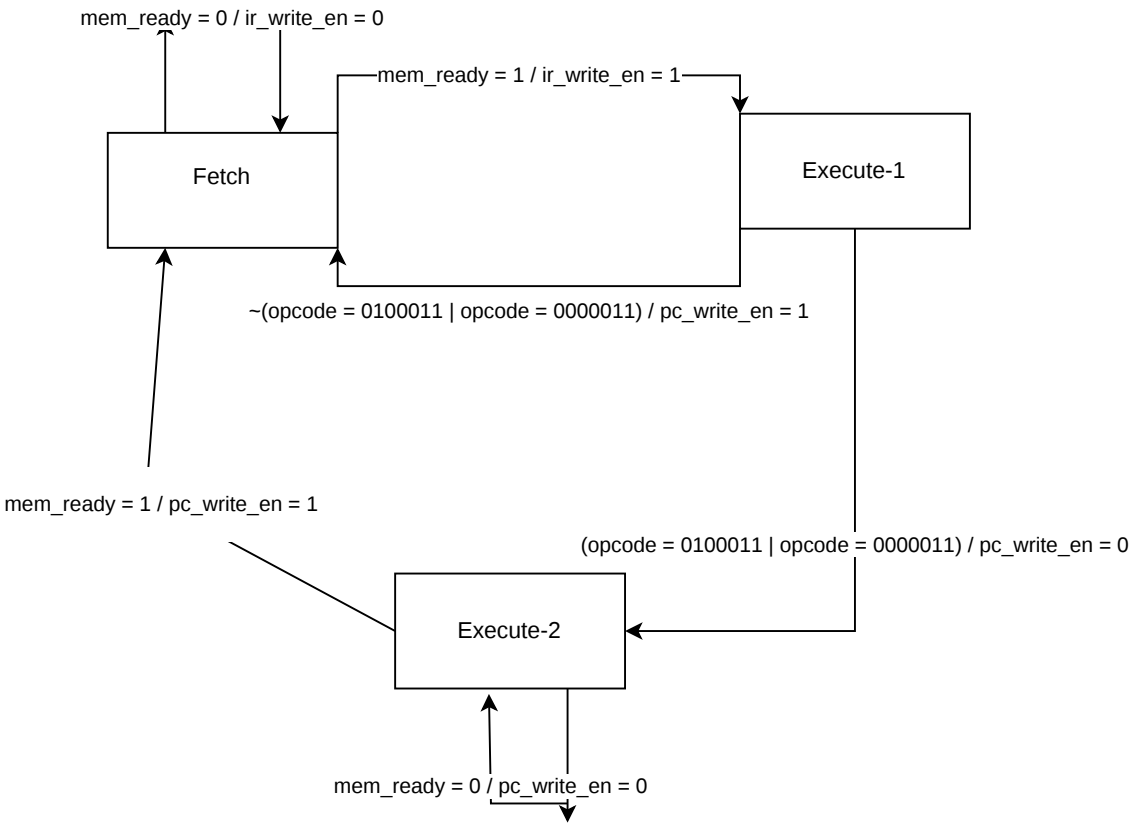
mem_ready = 0 / ir_write_en = 0
mem_ready = 1 / ir_write_en = 1
Fetch
Execute-1
~(opcode = 0100011 | opcode = 0000011) / pc_write_en = 1
mem_ready = 1 / pc_write_en = 1
(opcode = 0100011 | opcode = 0000011) / pc_write_en = 0
Execute-2
mem_ready = 0 / pc_write_en = 0